

Instagram Engagement Analysis Report

Project Title: Instagram Engagement Analysis and Content Strategy Recommendation for Alfido Tech

1. Introduction

Social media engagement is a key indicator of audience interest and brand performance. This project analyzes Instagram comment data to identify engagement trends, audience behavior, optimal posting schedules, and actionable strategies for Alfido Tech.

Objectives

- Analyze Instagram comment activity.
- Identify audience engagement patterns.
- Determine active posting days and times.
- Evaluate hashtag and emoji usage.
- Perform sentiment analysis.
- Recommend an optimal content calendar.

2. Dataset Description

Columns analyzed: id, comment, User id, Photo id, created Timestamp, posted date, emoji used, Hashtags used count.

3. Data Preprocessing

Removed duplicates, handled missing values, converted timestamps, and extracted Year, Month, Day, Hour, and Weekday features.

4. Exploratory Data Analysis

Audience activity by day and hour, most engaged posts, most active users, hashtag analysis, emoji analysis, comment length analysis, sentiment analysis, and monthly engagement trends.

5. Key Findings

Engagement varies by day and time. Certain posts generate more discussion. Emoji usage and hashtags influence interaction. Most audience sentiment is positive.

6. Recommended Content Calendar

Monday: Educational content

Tuesday: Product updates

Wednesday: Polls and interactive posts

Thursday: Behind-the-scenes content

Friday: Promotional content

Saturday: Community engagement

Sunday: Weekly recap

7. Strategies

1. Post during peak engagement hours.
2. Encourage participation through questions and polls.
3. Optimize hashtag usage.
4. Create community-focused content.
5. Replicate successful content themes.

8. Conclusion

The analysis provides actionable recommendations to improve engagement and strengthen Alfido Tech's social media presence.